



BIKE DEKHO - Bike Sales Analysis Excel Analysis

Project Overview:

The objective of this project is to analyze customer data to identify patterns and factors influencing bike purchases. Using Microsoft Excel, we explored various demographic and economic variables such as age, gender, marital status, income, education, and commute distance to uncover insights about customers' likelihood of purchasing a bike. The data was visualized using Pivot Tables, Pivot Charts, and interactive Dashboard. This analysis helps businesses like *Bike Dekho* make informed decisions in marketing, targeting, and product planning.

Project Objectives:

- ❖ To clean and organize raw bike sales data for better usability.
- ❖ To perform exploratory data analysis (EDA) using Excel formulas and PivotTables.
- ❖ To uncover key patterns in customer demographics, bike preferences, and sales performance.
- ❖ To create an interactive and insightful dashboard for decision-makers.

Key Tasks Performed:

1. Data Cleaning & Preprocessing:

- Replaced inconsistent values (e.g., changed “No” to “Not Purchased”).
- Formatted categorical variables (like gender, marital status).
- Converted data into structured, analysis-ready format.

2. Exploratory Data Analysis:

- a. Used Excel formulas to derive insights from demographics like:

- Age Group vs Bike Purchased
- Bike Purchase by Income Level
- Bike Purchase Distribution

b. Identified patterns and trends among target customer groups.

3. Data Visualization:

Created Pivot Tables and Pivot Charts to analyze:

- Bike Purchase Distribution
- Bike Purchase by Income Level
- Age Group vs Bike Purchased
- Bike Purchase by Commute Distance
- Marital Status vs Bike Purchase

4. Dashboard Creation:

- Designed a clean and insightful Excel dashboard.
- Consolidated charts from multiple analyses.
- Made it visually appealing and informative for quick decision-making.

Tools & Features Used:

- Microsoft Excel (Formulas, PivotTables, PivotCharts)
- Conditional Formatting
- Named Ranges and Data Validation
- Interactive Dashboard Design

Business Insights Generated:

- Nearly half of the potential customers convert, indicating a healthy interest that can be increased with targeted strategies.
- Focus marketing on middle-income groups (\$40K–\$60K), who form the majority of buyers.
- The middle-aged segment is the prime target for marketing and promotions.

- Short-distance commuters are more inclined to buy bikes, possibly for convenience and last-mile travel. Focus offers around short-distance travel benefits.
- Singles may be more independent in purchasing decisions or more in need of personal transport. Target singles with individual-centric offers.

Analysis Charts:

Chart 1

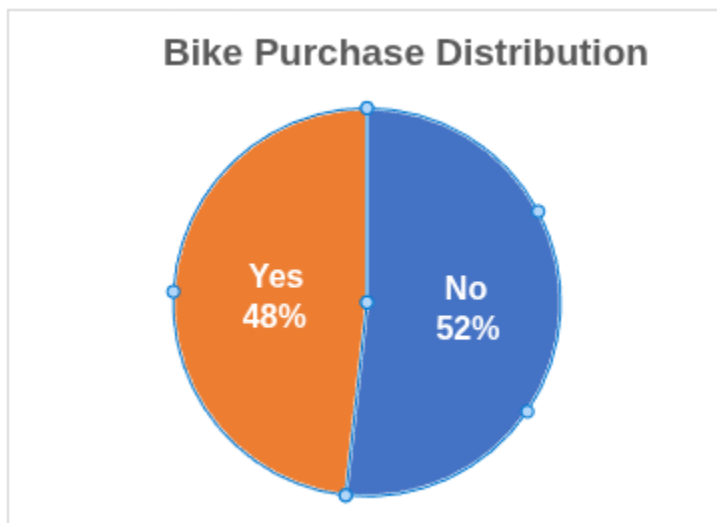


Chart 2

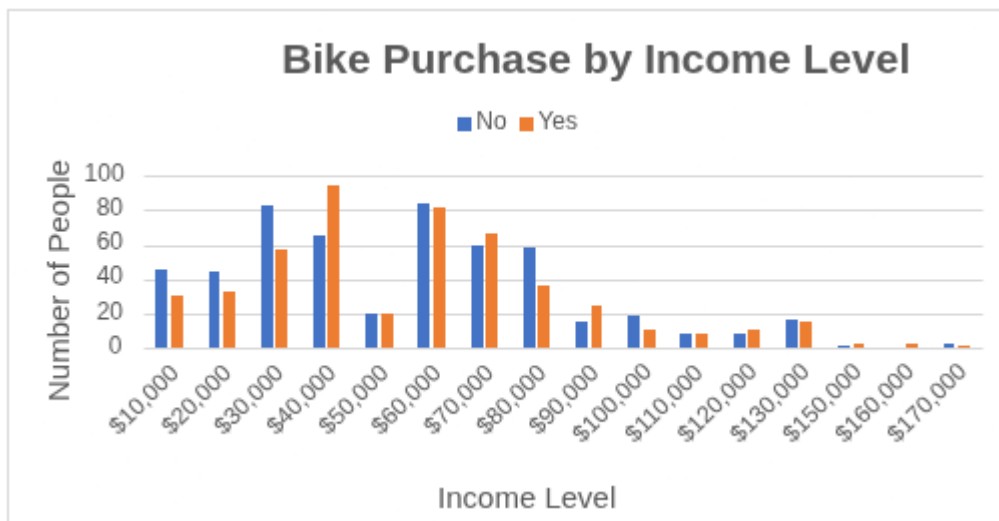


Chart 3

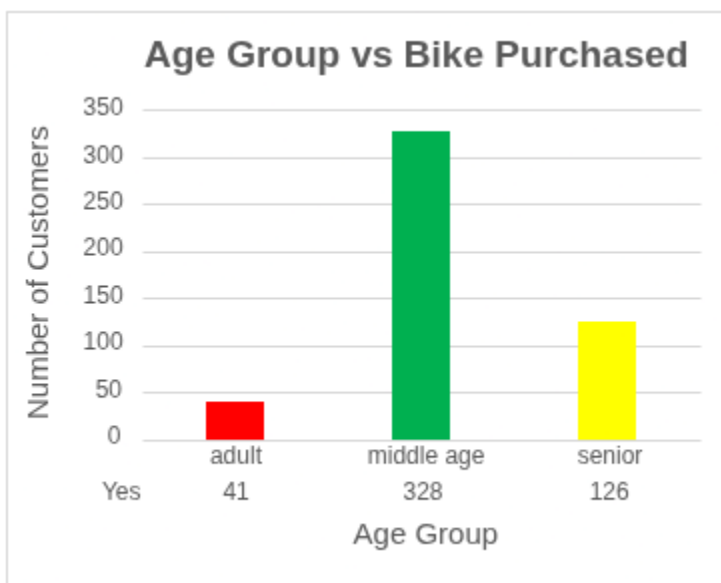


Chart 4

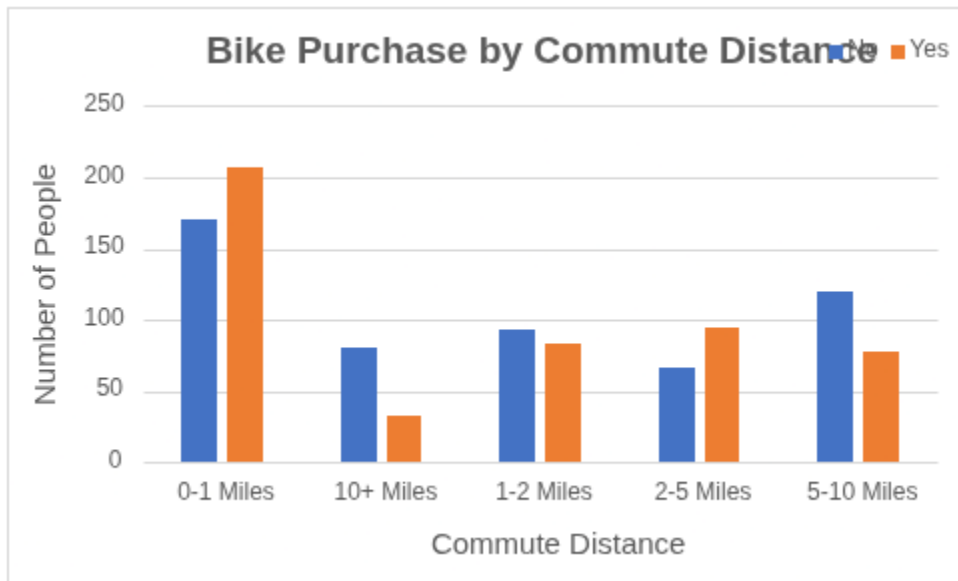


Chart 5

